

Edeviser

UI/UX Design System Review Document



Version 2.0 | April 12, 2026 | Living Document

OBE + Gamification Platform for Higher Education
Qatar Market | Arabic/English Bilingual

*Purpose: Comprehensive design reference for UI/UX review,
critique, and improvement recommendations.*

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Platform: OBE (Outcome-Based Education) + Gamification for Higher Education - Qatar market, Arabic/English bilingual.

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1. Design Philosophy & Principles

Core Principle: "Trust Through Clarity"

Edeviser serves two distinct user mindsets simultaneously:

- * Compliance-driven users (admins, coordinators, teachers) - need to complete administrative OBE tasks quickly and accurately.
- * Motivation-driven users (students) - need to feel progress, not just see numbers.

Five Design Laws

#	Law	Rationale
1	Solid over transparent	All data-bearing surfaces use solid back..
2	Hierarchy over decoration	Every visual element has a purpose. Deco..
3	Feedback is mandatory	Every user action produces an immediate ..
4	Brand consistency is non-negotiable	The teal-to-blue gradient is the brand f..
5	Mobile is first-class for students	Student views are designed mobile-first ..

Design Tone

- * Professional yet approachable - academic platform, not a social app.
- * Data-dense but not overwhelming - progressive disclosure, component-level loading.
- * Celebratory for achievements - gamification elements reward effort with visual delight.
- * Culturally respectful - full Arabic RTL support, no culturally insensitive imagery.

2. User Roles & Audience

Role	Primary Device	Key Tasks	Design Priority
Admin	Desktop	Institution setup, user manag..	Data density, efficiency
Coordinator	Desktop	Program management, curriculu..	Complex data visualization
Teacher	Desktop + Tablet	Grading, assignment creation,..	Workflow speed, grading UX
Student	Mobile + Desktop	Dashboard, assignments, journ..	Engagement, gamification, mob..
Parent	Mobile + Desktop	Child progress monitoring, at..	Read-only clarity, at-a-glanc..

3. Design Tokens & Color System

All visual values are defined as CSS custom properties in `src/index.css` and mapped to Tailwind CSS v4 via `@theme` blocks. No `tailwind.config.ts` - Tailwind v4 uses CSS-based configuration.

3.1 Brand Colors

Token	Hex Value	Tailwind Equivalent	Usage
<code>--brand-primary</code>	<code>#3b82f6</code>	<code>blue-500</code>	Primary CTA, active states, l..
<code>--brand-primary-dark</code>	<code>#2563eb</code>	<code>blue-600</code>	Hover states
<code>--brand-secondary</code>	<code>#14b8a6</code>	<code>teal-500</code>	Secondary accents, gradient s..
<code>--brand-secondary-dark</code>	<code>#0d9488</code>	<code>teal-600</code>	Teal hover states
<code>--gradient-start</code>	<code>#14b8a6</code>	-	Brand gradient start
<code>--gradient-end</code>	<code>#0382bd</code>	-	Brand gradient end

3.2 Brand Gradients (The Fingerprint)

The teal-to-blue gradient is the single most recognizable brand element. It appears on all card headers, primary buttons, and the logo.

`css

/ Standard Brand Gradient - card headers, primary buttons /

```
--brand-gradient: linear-gradient(93.65deg, #14B8A6 5.37%, #0382BD 78.89%);
```

/ Hero / Welcome Card Gradient - dashboard hero sections /

```
--hero-gradient: linear-gradient(135deg, #0f172a 0%, #1e3a8a 50%, #312e81 100%);
```

3.3 Semantic / Functional Colors

Token	Hex	Tailwind	Usage
--color-success	#22c55e	green-500	Completed states, pass grades
--color-warning	#f59e0b	amber-500	At-risk alerts, approaching d..
--color-destructive-brand	#ef4444		

ed-500 | Errors, delete actions, failed states |

--color-neutral	#64748b	slate-500	Secondary text, inactive elemen
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3.4 Surface Colors

Token	Hex	Tailwind	Usage
--surface-background	#ffffff	white	Page background
--surface-card	#ffffff	white	All card backgrounds
--surface-subtle	#f8fadc	slate-50	Alternate rows, inner contain..
--surface-border	#e2e8f0	slate-200	Card borders, dividers
--surface-input-border	#d1d5db	gray-300	Form input borders

3.5 Shadows

```
`css
```

```
--shadow-sm: 0 1px 2px 0 rgb(0 0 0 / 0.05);
```

```
--shadow-md: 0 4px 6px -1px rgb(0 0 0 / 0.1), 0 2px 4px -2px rgb(0 0 0 / 0.1);
```

```
--shadow-lg: 0 10px 15px -3px rgb(0 0 0 / 0.1), 0 4px 6px -4px rgb(0 0 0 / 0.1);
```

```
--shadow-xl: 0 20px 25px -5px rgb(0 0 0 / 0.1), 0 8px 10px -6px rgb(0 0 0 / 0.1);
```

3.6 Border Radius

```
`css
```

```
--radius: 0.625rem; / 10px base /
```

```
--radius-xl: calc(var(--radius) + 4px); / 14px - cards /
```

```
--radius-2xl: calc(var(--radius) + 8px); / 18px - hero cards /
```

3.7 Domain-Specific Color Coding

Bloom 2019s Taxonomy Levels

Each CLO has exactly one Bloom 2019s level. These colors are used consistently across pills, charts, and CLO tags:

Level	Color	Tailwind Class	Text
Remember	Purple	g-purple-500	ext-white
Understand	Blue	g-blue-500	ext-white
Apply	Green	g-green-500	ext-white
Analyze	Yellow	g-yellow-500	ext-gray-900
Evaluate	Orange	g-orange-500	ext-white
Create	Red	g-red-500	ext-white

Outcome Type Colors

Type	Background	Text	Border
ILO (Institutional)	g-red-100	ext-red-700	order-red-200
PLO (Program)	g-blue-100	ext-blue-700	order-blue-200
CLO (Course)	g-green-100	ext-green-700	order-green-200

Grade Attainment Colors

Level	Threshold	Text Color	Background
Excellent	>= 85%	ext-green-600	g-green-50
Satisfactory	70-84%	ext-blue-600	g-blue-50
Developing	50-69%	ext-yellow-600	g-yellow-50
Not Yet	< 50%	ext-red-600	g-red-50

Gamification Colors

Element	Color	Tailwind
XP / Gold	#f59e0b	mber-500
Streak Flame	gradient	

ed-500 to orange-500 |

Level Badge	Brand gradient	teal-to-blue
Leaderboard Gold	#fbf24	yellow-400
Leaderboard Silver	#9ca3af	gray-400
Leaderboard Bronze	#d97706	mber-600

4. Typography

Font Stack

Role	Family	Fallback
Primary (All UI)	Noto Sans	system-ui, sans-serif
Arabic (RTL)	Noto Sans Arabic	Noto Sans, system-ui, sans-serif
Code / Technical	Menlo	Monaco, Consolas, Courier New, monospace

Font loading: Google Fonts with display=swap for performance. Preconnect hint included.

Type Scale

Role	Tailwind Classes	Size	Weight	Usage
Page Title (H1)	ext-2xl font-bold track..	24px	700	Dashboard hero title
Section Heading (H2)	ext-lg font-bold tracki..	18px	700	Card titles, section he..
Subsection (H3)	ext-base font-semibold	16px	600	Widget titles, form gro..
KPI Value	ext-2xl font-black	24px	900	Metric numbers on stat ..
Metric Label	ext-[10px] font-black t..	10px	900	Labels beneath metric v..
Body	ext-sm font-medium anti..	14px	500	General content, descri..
Caption	ext-xs text-gray-500	12px	400	Helper text, timestamps
Navigation Pill	ext-sm font-medium	14px	500	Tab labels
Button	ext-sm font-semibold	14px	600	All button text
Badge Label	ext-xs font-bold tracki..	12px	700	Badge chips and tags
Code	ont-mono text-sm	14px	400	Code snippets, technica..

Typography Rules

- * Line height: Body text leading-relaxed (1.625); headings leading-tight (1.25).
- * Max width: Prose content areas max max-w-prose (65ch) for readability.
- * Antialiasing: Always apply ntialiased on text within cards.
- * Truncation: Long strings (names, titles) always use runcate - never wrap unexpectedly.

5. Spacing & Layout

Spacing Scale (Tailwind defaults - never use arbitrary values)

Token	Value	Use
p-2 / gap-2	8px	Icon padding, tight component gaps
p-3	12px	Small card padding
p-4	16px	Standard card padding
p-6	24px	Generous section padding
p-8	32px	Page-level padding
gap-4	16px	Standard grid/flex gap
gap-6	24px	Section-level gap

Grid System

- * Admin / Coordinator / Teacher: 12-column grid; use grid-cols-12 with col-span-*
- * Student: Responsive 3-column (grid-cols-1 md:grid-cols-2 lg:grid-cols-3).
- * KPI Cards Row: Always grid grid-cols-2 md:grid-cols-4 gap-4.

Page Layout Structure

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Top Nav Bar (h-16, fixed, bg-white, border-b)

+-----+-----+

Sidebar	Main Content Area
w-64	flex-1, p-6, bg-slate-50, space-y-6
bg-white	
border-r	[Hero Card]
	[KPI Row: 4 cards]
	[Tab Navigation]
	[Tab Content / Section Cards]

+-----+-----+

* Sidebar: w-64 fixed width, g-white, order-r border-slate-200

* Main content:

flex-1 overflow-auto p-6 bg-slate-50

* Student mobile: sidebar hidden, bottom nav bar instead

6. Iconography

Library: Lucide React (tree-shakeable, 2px stroke, rounded caps/joins)

Size Rules

Context	Size	Class
Navigation sidebar	20x20px	h-5 w-5
Inside buttons	16x16px	h-4 w-4 with gap-2
KPI card icons	20x20px	h-5 w-5 inside gradient container
Empty state	40x40px	h-10 w-10
Decorative (hero bg)	64x64px	h-16 w-16 opacity-10

Semantic Icon Map

Concept	Icon Name	Context
Dashboard	LayoutDashboard	Navigation
Students/Users	Users	Navigation, headings
Teachers	GraduationCap	Navigation, badges
ILO / Outcomes	Target	OBE section
PLO	Layers	Coordinator section
CLO	BookOpen	Teacher section
Assignment	FileText	Assignment management
Rubric	Grid3X3	Rubric builder
Evidence	ShieldCheck	Evidence/compliance
Report / Export	Download	Report actions
Analytics	BarChart3	Dashboard sections

XP	Zap	XP displays
Streak	Flame	Streak counter
Badge	Award	Badge display
Level	Star	Level indicator
Leaderboard	Trophy	Leaderboard section
AI	Sparkles	AI Co-Pilot features
Journal	NotebookPen	Reflection journal
Warning / At-Risk	AlertTriangle	Warning states
Add / Create	Plus	Creation actions
Edit	Pencil	Edit actions
Delete	Trash2	Destructive actions
Settings	Settings	Config sections

Icon Rules

- * Never use icons alone as the only visual affordance - always pair with a label or tooltip.
- * All icon-only buttons must have a label.

7. Component Library

Built on Shadcn/ui (New York style, Radix primitives). All interactive elements MUST use Shadcn/ui - never raw HTML.

Architecture

- * Shadcn/ui primitives: src/components/ui/ (Button, Card, Dialog, Form, Input, Select, Tabs, Badge, etc.)
- * Custom shared components: src/components/shared/ (DataTable, GradientCardHeader, StreakDisplay, BadgeCollection, etc.)
- * CVA (class-variance-authority) for component variants
- * cn() utility from src/lib/utils.ts for merging Tailwind classes

7.1 Buttons

Variant	Style	Usage	Max per Section
Primary (Gradient CTA)	Teal-to-blue gradient, white ..	Main action	1
Secondary	Outline, border-gray-200, bg-..	Cancel, secondary actions	Unlimited
Destructive	bg-red-500, hover:bg-red-600	Delete, remove	As needed
Ghost (Icon)	Ghost variant, text-gray-500,..	Inline edit/delete	As needed

Button rules:

- * Minimum touch target: 44x44px on mobile
- * Loading state: replace content with Loader2 spinner (animate-spin)
- * Press feedback: active:scale-95 transition-transform duration-100

7.2 Cards

Card Type	Style	When to Use
Standard Data Card	bg-white border-0 shadow-md hover:shadow..	General content containers
Gradient Header Card	Card with overflow-hidden, header div wi..	ALL main section cards on dashboards
KPI Metric Card	group bg-white border-0 shadow-lg rounde..	Stat/metric displays
Hero Welcome Card	Dark gradient (135deg, #0f172a, #1e3a8a,..	Top of all role dashboards

Card rules:

- * All cards: bg-white border-0 shadow-md rounded-xl
- * Section card headers use inline style for the brand gradient (not Tailwind class)
- * Card must have overflow-hidden for gradient to clip to rounded corners
- * KPI cards use group hover with icon scale-110 transition

7.3 Tab Navigation (Pill Row)

State	Style
Container	flex w-full gap-2 (no connected bar)
Active	bg-blue-600 text-white shadow-md border-blue-600 - NEVER grad..
Inactive	bg-white text-gray-600 border-gray-200 shadow-sm hover:bg-gra..
Shape	rounded-xl with icon + label

Critical: Never use pink/purple/violet as tab active colors.

7.4 Tables / Data Tables

- * Wrapper: rounded-xl overflow-hidden border border-gray-200 bg-white shadow-sm
- * Header row: bg-gray-50 border-b border-gray-200
- * Header text: text-xs font-black text-gray-500 tracking-widest uppercase
- * Alternating rows: bg-white / bg-gray-50/30
- * Hover: hover:bg-slate-50 transition-colors
- * Action buttons: ghost icon buttons with blue (edit) and red (delete) hover states
- * Always use the shared DataTable component with TanStack Table

7.5 Select / Dropdown

- * Trigger: bg-white border border-gray-200 rounded-lg shadow-sm
- * Content: bg-white border border-gray-200 shadow-lg rounded-xl
- * Items: rounded-lg hover:bg-blue-50 cursor-pointer
- * CRITICAL: Never bg-transparent or bg-gray-900 for SelectTrigger

7.6 Loading States

- * Use component-level shimmer skeletons, NEVER full-page blocking loaders
- * Shimmer: gradient sweep animation (animate-shimmer)
- * Pattern: from-gray-200 via-gray-100 to-gray-200 bg-[length:200% 100%]
- * Place shimmer placeholders matching the shape of the content they replace

7.7 Empty States

- * Centered layout: flex flex-col items-center justify-center py-16 text-center
- * Icon in muted container: p-4 bg-gray-100 rounded-2xl

- * Heading + description + CTA button
- * CTA uses gradient primary button style

7.8 Key Shared Components

Component	Path	When to Use
DataTable	src/components/shared/DataTable.tsx	Any data table with sort/filter/pagination
GradientCardHeader	src/components/shared/GradientCardHeader..	Section card headers
ConfirmDialog	src/components/shared/ConfirmDialog.tsx	Destructive action confirmation
ErrorBoundary	src/components/shared/ErrorBoundary.tsx	Component error boundaries
BadgeCollection	src/components/shared/BadgeCollection.tsx	Badge displays
StreakDisplay	src/components/shared/StreakDisplay.tsx	Streak counters
LevelProgress	src/components/shared/LevelProgress.tsx	XP level bars
XPAwardToast	src/components/shared/XPAwardToast.tsx	XP earned feedback
BadgeAwardModal	src/components/shared/BadgeAwardModal.tsx	Badge unlock celebration
LevelUpOverlay	src/components/shared/LevelUpOverlay.tsx	Level up full-screen celebration
MysteryBadge	src/components/shared/MysteryBadge.tsx	Hidden badge silhouettes
BonusEventBanner	src/components/shared/BonusEventBanner.tsx	Active XP multiplier events
HabitTracker	src/components/shared/HabitTracker.tsx	7-day habit grid
BloomsVerbGuide	src/components/shared/BloomsVerbGuide.tsx	CLO creation helper
RubricPreview	src/components/shared/RubricPreview.tsx	Rubric display
CurriculumMatrix	src/components/shared/CurriculumMatrix.tsx	PLO x Course grid
RealtimeStatusBanner	src/components/shared/RealtimeStatusBann..	Live update status

8. Page Patterns & Templates

Every page follows one of these standardized patterns:

8.1 List Page Pattern

Used for: User List, Program List, Course List, ILO/PLO/CLO List, Assignment List, etc.

Structure:

1. Page header: title (text-2xl font-bold tracking-tight) + gradient CTA button
2. Filter bar: search input with icon, dropdown filters (nuqs for URL-persisted state)
3. DataTable component with TanStack Table (sort, filter, pagination)

Key rules:

- * Always use nuqs for URL-persisted search/filter state
- * Always use TanStack Table via the shared DataTable component
- * One gradient CTA button max per page header
- * Loading state handled by isLoading from TanStack Query

8.2 Form Page Pattern

Used for: Create/Edit User, Program, Course, ILO, PLO, CLO, Assignment, etc.

Structure:

1. Card wrapper: bg-white border-0 shadow-md rounded-xl p-6 max-w-2xl

2. React Hook Form + Zod resolver for validation
3. Shadcn Form components (FormField, FormItem, FormLabel, FormControl, FormMessage)
4. Gradient submit button with Loader2 spinner when pending
5. Sonner toast for success/error feedback

8.3 Dashboard Page Pattern

Used for: Admin, Coordinator, Teacher, Student dashboards

Structure:

1. Welcome Hero Card (dark gradient, user greeting)
2. KPI Row: grid grid-cols-2 md:grid-cols-4 gap-4
3. Tab Navigation (pill style)
4. Section Cards with gradient headers

8.4 Layout Pattern (Sidebar)

Used for: Admin, Coordinator, Teacher layouts

- * Sidebar: w-64 fixed, bg-white, border-r border-slate-200
- * Active nav item: bg-blue-50 text-blue-600
- * Inactive nav item: text-gray-600 hover:bg-slate-50
- * Nav icons: h-5 w-5
- * Main content area: flex-1 overflow-auto p-6 bg-slate-50

9. Dashboard Patterns

9.1 Welcome / Hero Card

Appears at the top of ALL role dashboards. Contains user greeting and high-level KPI summary.

- * Background: hero gradient (135deg, #0f172a -> #1e3a8a -> #312e81)
- * Optional texture overlay at 5% opacity
- * Dark overlay (bg-black/20) for text readability
- * Content: greeting (text-2xl font-bold text-white), subtitle (text-blue-200 text-sm)
- * Bottom row: inline KPI stats (label in text-blue-300 text-xs uppercase, value in text-white text-2xl font-black)

9.2 Section Card Header (The Brand Fingerprint)

Every main category section card uses this exact background on its header:

```
background: linear-gradient(93.65deg, #14B8A6 5.37%, #0382BD 78.89%)
```

With optional subtle educational pattern overlay at 10% opacity.

Rules:

- * Used on EVERY main section card across ALL role dashboards. No exceptions.
- * Header: px-6 py-4, white icon (h-5 w-5), white heading text (text-lg font-bold tracking-tight)
- * Optional subtitle in text-blue-100 text-sm
- * Card body: p-6 on white background

9.3 KPI Metric Card

- * Container: group bg-white border-0 shadow-lg hover:shadow-xl rounded-xl overflow-hidden
- * Layout: flex row, metric info left, icon container right
- * Metric title: text-sm font-bold text-gray-800
- * Metric value: text-2xl font-black text-gray-900 (hover: text-blue-700)
- * Icon container: p-2 bg-gradient-to-br from-blue-400/20 to-blue-600/20 rounded-xl
- * Hover: icon container scale-110 rotate-3 transition
- * Bottom accent line: h-0.5 gradient from-blue-400 to-teal-500, opacity-0 -> opacity-100 on hover

9.4 Attainment Progress Bar

- * Label + percentage on same line (justify-between)
- * Track: h-2 bg-gray-100 rounded-full overflow-hidden
- * Fill: gradient based on attainment level (green >= 70%, yellow >= 50%, red < 50%)
- * Animated width transition: transition-all duration-700 ease-out

10. Gamification UI Patterns

10.1 XP Display (Header Chip)

- * Container: bg-amber-50 border border-amber-200 rounded-full px-3 py-1
- * Icon: Zap h-3.5 w-3.5 text-amber-500
- * Text: text-sm font-black text-amber-700

10.2 Streak Display

- * Container: bg-gradient-to-r from-orange-50 to-red-50 border border-orange-200 rounded-xl p-3
- * Icon: Flame h-6 w-6 text-orange-500 animate-streak-flame
- * Value: text-xl font-black text-gray-900
- * Label: text-[10px] font-black tracking-widest uppercase text-orange-600

10.3 Badge Display

Compact chip:

- * Container: bg-purple-50 border border-purple-200 rounded-full px-2.5 py-1
- * Emoji + text-xs font-bold text-purple-700

Full card:

- * Container: bg-white border border-gray-200 rounded-xl shadow-sm hover:shadow-md p-4 text-center
- * Emoji: text-4xl with group-hover:animate-badge-pop
- * Name: text-xs font-bold text-gray-800
- * Date: text-[10px] text-gray-400

Mystery badge (unearned):

- * Same card but emoji has opacity-30 grayscale
- * Name shows "???"
- * On unlock: animate-badge-pop + confetti burst

10.4 Level Progress Bar

- * Label: "Level N - Title" (text-sm font-bold text-gray-700)

- * XP counter: "900 / 1,400 XP" (text-xs text-gray-500)
- * Track: h-3 bg-gray-100 rounded-full
- * Fill: bg-gradient-to-r from-teal-400 to-blue-600, transition-all duration-1000
- * Remaining: text-[10px] text-gray-400 "500 XP to Level 6"

10.5 Leaderboard Row

- * Rank badge: w-6 h-6 rounded-full, gold/silver/bronze/gray based on position
- * Avatar: w-8 h-8 rounded-full gradient (blue-400 to teal-500) with initial
- * Name + level info
- * XP value: text-sm font-black text-amber-600

10.6 Bonus Event Banner

- * Background: brand gradient
- * Icon in white/20 container
- * Event title + multiplier description
- * Countdown timer on right side

11. AI Co-Pilot UI Patterns

11.1 AI Suggestion Card (Student Dashboard)

- * Left border accent: border-l-4 border-l-blue-400
- * Sparkles icon in bg-blue-50 container
- * "Suggested for you" heading + "AI Draft" italic label
- * Suggestion text + supporting stat
- * Thumbs up/down feedback buttons at bottom
- * "Was this helpful?" prompt

11.2 At-Risk Student Row (Teacher Dashboard)

- * Background: bg-red-50 with border-l-4 border-l-red-400
- * Red avatar with student initial
- * Student name + last login + risk reason
- * Probability badge: bg-red-100 text-red-700 border border-red-200
- * "Send Nudge" action button

11.3 AI Feedback Draft (Teacher Grading View)

- * Background: bg-amber-50 border border-amber-200 rounded-lg
- * Sparkles icon + "AI Draft" italic label
- * Draft comment text
- * Three action buttons: Accept (green), Edit (outline), Reject (red ghost)

11.4 AI-Generated Content Label

All AI-generated content MUST include this transparency label:

- * Style: text-xs text-gray-400 italic
- * Text: "AI Draft"

12. Self-Regulated Learning UI

12.1 CLO Progress Bar

- * Bloom's level pill (color-coded per taxonomy)
- * CLO title (truncated)
- * Attainment percentage badge (color-coded per attainment level)
- * Progress bar with attainment-level gradient fill

12.2 XP Transaction Row

- * Icon in colored circle (green for positive, red for negative XP)
- * Source label + relative timestamp
- * Amount: font-black, green for positive, red for negative

12.3 Habit Tracker (7-Day Grid)

- * 7 columns (one per day), 4 rows (Login, Submit, Journal, Read)
- * Day label: text-[10px] text-gray-400
- * Completed: w-4 h-4 rounded-sm bg-green-500
- * Incomplete: w-4 h-4 rounded-sm bg-gray-200

12.4 Learning Path

- * Visual node-based path showing assignments ordered by Bloom's level
- * Locked nodes: grayed out with lock icon
- * Unlocked nodes: colored by Bloom's level
- * Completed nodes: checkmark overlay
- * Node unlock animation: animate-node-unlock

13. Notification Patterns

13.1 Peer Milestone Toast

- * Sonner custom toast layout
- * Star icon in bg-blue-50 container
- * "Peer Milestone!" heading with emoji
- * Classmate name + achievement description

13.2 Perfect Day Prompt

- * Background: bg-amber-50 border border-amber-200 rounded-xl
- * Sparkle emoji
- * "Almost a Perfect Day!" heading
- * Missing habit highlighted in font-bold text-amber-700
- * "Do it now" gradient CTA button

13.3 Toast Notifications (General)

- * Library: Sonner
- * Success: toast.success()
- * Error: toast.error()
- * Custom: toast.custom() for rich layouts
- * Position: top-right (default)

14. Form & Input Patterns

Standard Form Field

- * Label: text-sm font-semibold text-gray-700
- * Required indicator: red asterisk after label
- * Input: bg-white border-gray-200 focus:border-blue-500 focus:ring-2 focus:ring-blue-500/20 rounded-lg
- * Helper text: text-xs text-gray-400
- * Error state: border-red-400 bg-red-50 + red helper text
- * Success state: border-green-400
- * Disabled state: opacity-50 cursor-not-allowed

Form Layout Rules

- * Field spacing: space-y-4 between field groups
- * Submit button: always at bottom, gradient primary style
- * Full width submit on mobile sheets
- * Always use React Hook Form + Zod resolver
- * Always use Shadcn Form components

15. Animation & Motion

Motion Philosophy

- * Purposeful: Every animation communicates something (progress, success, state change). No animation for decoration alone.
- * Fast: Micro-interactions $\leq 200\text{ms}$. Celebrations $\leq 600\text{ms}$. Never block user action.
- * Respectful: Honor prefers-reduced-motion. All animations disabled for users with this preference.

Standard Easing Functions

Name	Value	Usage
ease-out-expo	cubic-bezier(0.16, 1, 0.3, 1)	Snappy exits, overlays
ease-in-out	cubic-bezier(0.4, 0, 0.2, 1)	State transitions
ease-spring	cubic-bezier(0.34, 1.56, 0.64, 1)	Badge pop, level up

Animation Catalog

Name	Duration	Trigger	Type
fade-in	200ms	Component mount	Framer Motion
slide-in-up	300ms	Card mount	Framer Motion
shimmer	1.5s loop	Loading state	CSS keyframes
badge-pop	500ms	Badge award	CSS keyframes (spring)
xp-pulse	600ms	XP earned	CSS keyframes
xp-count-up	800ms	XP earned	JS counter
streak-flame	800ms loop	Active streak	CSS keyframes
level-up	600ms	Level up event	Framer Motion overlay
confetti	2s	Submission/achievement	canvas-confetti library
float	3s loop	Background decorations	CSS keyframes
button-press	100ms	Button click	CSS active:scale-95
node-unlock	500ms	Learning path unlock	CSS keyframes
mystery-reveal	600ms	Mystery badge reveal	CSS keyframes (spring)
fade-in-up	300ms	Content entrance	CSS keyframes

Reduced Motion Support

All custom animations are disabled via:

```
@media (prefers-reduced-motion: reduce) {
  .animate-shimmer, .animate-xp-pulse, .animate-badge-pop,
  .animate-float, .animate-streak-flame, .animate-fade-in-up,
  .animate-node-unlock, .animate-mystery-reveal {
    animation: none;
  }
}
```

Additionally, a `.reduce-animations` CSS class is available for user-controlled animation preferences.

16. Responsive Design

Breakpoints (Tailwind defaults)

Prefix	Min-width	Target
(none)	0px	Mobile: 360-767px
sm:	640px	Large mobile / small tablet
md:	768px	Tablet
lg:	1024px	Desktop
xl:	1280px	Wide desktop

Responsive Behavior by Role

View	Mobile	Tablet	Desktop
Student Dashboard	Single column, bottom nav	2-col cards	3-col cards, sidebar
Teacher Dashboard	Tabs stacked, grading full-sc..	2-col	3-col, split grading view
Admin/Coordinator	Limited access notice (or sim..	2-col	Full 3-col + sidebar

Student Mobile Navigation

On mobile (below md: breakpoint), the sidebar is replaced with a fixed bottom navigation bar:

- * Position: fixed bottom-0, full width
- * Background: bg-white border-t border-gray-200
- * Layout: flex justify-around py-2
- * Items: icon (h-5 w-5) + label (text-xs)
- * z-index: z-50
- * Hidden on md: and above (md:hidden)

17. RTL & Internationalization

Languages

- * English (LTR) - primary
- * Arabic (RTL) - full support
- * i18next + react-i18next for language switching

RTL Implementation

- * HTML dir attribute: <html dir="rtl"> when Arabic is active
- * Arabic font: "Noto Sans Arabic" loaded via [dir="rtl"] CSS selector
- * Logical CSS properties used throughout:
 - * ms- / me- instead of ml- / mr- (margin)
 - * ps- / pe- instead of pl- / pr- (padding)
 - * start / end instead of left / right for positioning
- * Radix UI popovers respect direction via CSS override
- * Locale files: public/locales/{ar,ur}/ and src/locales/{en,ar}/

Translation File Structure

- * Namespaced by role: auth.json, admin.json, teacher.json, student.json, common.json
- * Keys follow dot notation: admin.users.createButton
- * All user-facing strings must be in translation files - no hardcoded English

18. Accessibility Standards

Target: WCAG 2.1 Level AA compliance on all core learning flows.

Color Contrast Audit

Token	Hex	Ratio vs White	WCAG AA Status
--brand-primary (blue-500)	#3b82f6	3.13:1	Large text only
--brand-primary-dark (blue-600)	#2563eb	4.57:1	All text
--brand-secondary (teal-500)	#14b8a6	2.90:1	Large text/UI only
--color-success (green-500)	#22c55e	2.52:1	Use on bg-*-50 only
--color-warning (amber-500)	#f59e0b	2.15:1	Use on bg-*-50 only
--color-destructive (red-500)	#ef4444	4.00:1	Large text
--color-neutral (slate-500)	#64748b	4.63:1	All text
text-gray-500	#6b7280	4.64:1	All text
text-gray-900 (body)	#111827	17.4:1	All text

Guidelines:

- * Use --brand-primary-dark (#2563eb) for interactive text links (not blue-500)
- * Success/warning colors MUST be paired with tinted backgrounds (bg-green-50, bg-yellow-50)
- * Never rely on color alone to convey information - always pair with text labels, icons, or patterns

Attainment Badge Contrast (all pass AA)

- * Excellent: text-green-600 on bg-green-50 (7.08:1)
- * Satisfactory: text-blue-600 on bg-blue-50 (6.20:1)
- * Developing: text-yellow-600 on bg-yellow-50 (4.68:1)
- * Not Yet: text-red-600 on bg-red-50 (6.07:1)

Accessibility Checklist

Category	Requirement	Implementation
Color Contrast	Body text $\geq$ 4.5:1, Large text $\geq$ 3:1	Verified per audit table above
Focus Indicators	Visible focus ring on all interactive el..	focus:ring-2 focus:ring-blue-500 focus:r..
Keyboard Navigation	All interactions via keyboard	Tab order logical; Esc closes modals; En..
ARIA Labels	Meaningful labels for icon-only buttons	aria-label on all icon buttons
Alt Text	All meaningful images have alt text	alt="" for decorative images
Form Labels	All inputs have associated labels	Label htmlFor or aria-label
Error Messages	Errors announced to screen readers	role="alert", aria-invalid="true"
Skip Navigation	Skip-to-main-content link	sr-only focus:not-sr-only at page top
Motion	Reduced motion respected	@media (prefers-reduced-motion: reduce)

Additional Accessibility Features (Implemented)

- * Dyslexia-friendly font option: .dyslexia-font class (OpenDyslexic)
- * High contrast mode: .high-contrast class (black borders, black text)
- * Font size overrides: html.font-large (18px), html.font-x-large (20px)
- * Simplified view: .simplified-view hides non-essential decorative elements

19. Dark Mode

Current Status: Phase 2 (Student dashboard first)

Implementation

- * Strategy: CSS class .dark on <html> element
- * ThemeProvider: src/providers/ThemeProvider.tsx
- * Toggle: user preference stored in localStorage

Dark Mode Token Overrides

Token	Light Value	Dark Value
--surface-background	#ffffff	#020617 (slate-950)
--surface-card	#ffffff	#0f172a (slate-900)
--surface-subtle	#f8fafc	#1e293b (slate-800)
--surface-border	#e2e8f0	#334155 (slate-700)
--surface-input-border	#d1d5db	#475569 (slate-600)
--text-primary	#0f172a	#f1f5f9 (slate-100)
--text-secondary	#64748b	#94a3b8 (slate-400)

Shadows are intensified in dark mode (0.2-0.3 opacity vs 0.05-0.1 in light).

Shimmer animation uses slate-800/slate-700 gradient in dark mode.

20. Prohibited Patterns

These patterns are HARD RULES - never use them:

Pattern	Why Prohibited	Permitted Alternative
Pink, Purple, Violet, Rose, Fuchsia back..	Off-brand; conflicts with brand palette	Blue (#3b82f6), Teal (#14b8a6)
Transparent/glassmorphism on data cards	Readability fails on varied backgrounds	Solid bg-white or bg-gray-50
Transparent SelectTrigger backgrounds	Input appears invisible on light surfaces	bg-white border border-gray-200
Unicode bullet characters in code	Inconsistent rendering	Semantic ul/li elements
Full-page skeleton loaders	Perceived performance loss	Optimistic UI + component-level shimmer
Raw HTML for interactive elements	Inconsistent styling, no accessibility	Shadcn/ui components
Arbitrary spacing values	Inconsistent layout	Tailwind spacing scale
Direct Supabase calls in components	Breaks architecture pattern	TanStack Query hooks
any TypeScript type	Type safety violation	unknown with type guards, proper interfa..
Hardcoded English strings	Breaks i18n	Translation keys via i18next
Physical CSS properties (ml, mr, pl, pr)	Breaks RTL	Logical properties (ms, me, ps, pe)

21. Implementation Checklist

Use this checklist when building any new dashboard view or component:

Layout & Structure

- [] Page uses bg-slate-50 background
- [] Cards use bg-white border-0 shadow-md rounded-xl

- [] Main section cards use the brand gradient on header
- [] KPI metric row uses grid grid-cols-2 md:grid-cols-4 gap-4
- [] Sidebar (if present) is bg-white border-r border-slate-200

Typography

- [] Page title is text-2xl font-bold tracking-tight
- [] Section headings are text-lg font-bold tracking-tight
- [] Metric values are text-2xl font-black
- [] Metric labels are text-[10px] font-black tracking-widest uppercase
- [] Body text uses text-sm font-medium antialiased

Colors & Brand

- [] No pink, purple, violet, rose, or fuchsia on cards or tabs
- [] Active tabs use bg-blue-600 text-white (never gradient)
- [] Select triggers use bg-white border border-gray-200
- [] No glassmorphism on data cards
- [] Brand gradient used on all section card headers

Interactions

- [] Buttons have active:scale-95 press state
- [] Hover states defined and don't cause icons to vanish
- [] Edit icon hovers text-blue-600 bg-blue-50
- [] Delete icon hovers text-red-600 bg-red-50
- [] Loading states use shimmer, not spinners alone

Accessibility

- [] All icon-only buttons have aria-label
- [] Focus rings present on all interactive elements (focus:ring-2 focus:ring-blue-500)
- [] Color contrast verified for text elements
- [] Form inputs have associated labels
- [] Error messages use role="alert"
- [] Skip navigation link present
- [] prefers-reduced-motion respected

Gamification (Student views)

- [] XP displayed in amber chip in header
- [] Streak displayed with flame icon and animation
- [] Level progress bar present on profile/hero card
- [] Badge awards trigger animate-badge-pop
- [] Mystery badges show as silhouettes (opacity-30 grayscale) until earned
- [] Bonus Event banner uses brand gradient with countdown timer

AI Co-Pilot

- [] AI suggestion cards use border-l-4 border-l-blue-400 with sparkles icon
- [] At-risk student rows use bg-red-50 border-l-4 border-l-red-400
- [] Feedback drafts use bg-amber-50 border border-amber-200
- [] All AI-generated content includes "AI Draft" label (text-xs text-gray-400 italic)
- [] Thumbs up/down feedback buttons on all AI suggestions

Self-Regulated Learning

- ☐ CLO Progress bars include Bloom's level pill (color-coded)
- ☐ CLO attainment colors follow design system (green/blue/yellow/red)
- ☐ XP Transaction rows show positive in green, negative in red
- ☐ Habit Tracker uses 7-day grid with green/gray squares
- ☐ Peer milestone notifications via Sonner toast

RTL / i18n

- ☐ Logical CSS properties used (ms/me/ps/pe, not ml/mr/pl/pr)
- ☐ All user-facing strings in translation files
- ☐ Arabic font loads correctly in RTL mode
- ☐ Layout mirrors correctly in RTL

22. Open Questions for UI/UX Review

The following areas are flagged for your constructive criticism and improvement suggestions:

Brand & Visual Identity

1. Is the teal-to-blue gradient distinctive enough as a brand fingerprint, or does it feel generic?
2. Does the hero card dark gradient (slate-900 -> blue-800 -> violet-900) create sufficient visual hierarchy?
3. Are we over-relying on the brand gradient? Should some section cards use alternative header treatments?
4. Is the color palette sufficiently differentiated for users with color vision deficiencies?

Information Architecture

5. Is the sidebar navigation structure intuitive for each role? Are there too many items?
6. Does the KPI card row (4 cards) surface the right metrics for each role's primary tasks?
7. Is the tab navigation pattern (pill style) discoverable enough, or should we consider a different pattern?
8. Are empty states providing enough guidance to help users take their first action?

Gamification UX

9. Is the XP/streak/badge system visually overwhelming or does it enhance the learning experience?
10. Are mystery badges (silhouettes) motivating or frustrating for students?
11. Does the leaderboard design adequately handle the anonymous opt-out case?
12. Is the "Perfect Day" nudge notification timing (6 PM) appropriate, or could it feel intrusive?
13. Are celebration animations (confetti, badge-pop, level-up overlay) too much or just right?

AI Co-Pilot UX

14. Is the "AI Draft" label sufficient for transparency, or do we need more prominent disclosure?
15. Are the thumbs up/down feedback buttons discoverable enough on AI suggestion cards?
16. Does the at-risk student row design create appropriate urgency without causing alarm?
17. Should AI feedback drafts in the grading view be collapsed by default?

Accessibility & Inclusion

18. Are the current contrast ratios sufficient? Several brand colors (teal-500, green-500) only pass for large text.

- 19. Is the dyslexia font option discoverable enough in the settings?
- 20. Should we offer a "simplified view" toggle more prominently for users who find the gamification distracting?
- 21. Are the animation durations appropriate for users with vestibular disorders (even with reduced motion)?

Mobile Experience

- 22. Is the bottom navigation bar for students sufficient, or do we need a hamburger menu for secondary items?
- 23. How should the grading interface adapt for tablet use by teachers?
- 24. Are touch targets consistently 44x44px minimum across all interactive elements?
- 25. Should the habit tracker grid be replaced with a different visualization on small screens?

Dark Mode

- 26. Is the current dark mode token set sufficient, or are there contrast issues in dark mode?
- 27. Should dark mode be available for all roles or just students (current plan)?
- 28. How should the brand gradient render in dark mode - same values or adjusted?

Data Visualization

- 29. Are Recharts the right choice for our chart needs, or should we consider a more accessible charting library?
- 30. Is the curriculum matrix (PLO x Course grid) usable at scale (20+ PLOs, 30+ courses)?
- 31. Does the Sankey diagram for outcome flow provide actionable insights or just visual complexity?

Cultural Considerations

- 32. Are there any visual elements that may not translate well to the Qatar/Gulf cultural context?
- 33. Is the emoji usage in badges and notifications appropriate for an academic setting?
- 34. Should the greeting pattern ("Good morning, Name") adapt to local customs?

This document consolidates all design decisions currently implemented in Edeviser. It is intended as a complete reference for UI/UX review. When in doubt about any pattern: solid white backgrounds, blue-teal gradient headers, no purple tabs.

Source files: [DESIGN-STYLE-GUIDE.md](#), [.kiro/steering/design-system.md](#), [.kiro/steering/component-patterns.md](#), [.kiro/steering/figma-design-system-rules.md](#), [src/index.css](#)